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Browsing the World in the Sensors' Continuum

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Motivations

- Computer-based systems and sensors will be soon embedded in everywhere
  - all our everyday objects
  - all our everyday environments
- Current deployments of pervasive services and sensor networks focus on special purpose systems, e.g.,
  - E.g., environmental monitoring and healthcare
- General purpose approaches are likely to emerge soon
  - Shared infrastructures of sensors, tags, smart-objects, cameras, Web 2.0 data
  - Generating general-purpose data ("facts") about the physical and the social worlds
  - Defining a sort of distributed digital "***world model***"
  - For general-purpose exploitation by "**browsing the world**" services



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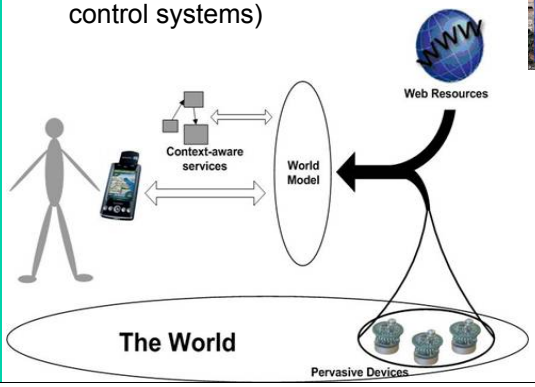
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"Browsing the World" Scenario

Exploit the world model to

- Implement general-purpose services to help us interact with the physical world (e.g., personalized real-time maps)
- Have services coordinate with each other in a context-aware fashion to achieve global goals (e.g., traffic control systems)





PLEASE NOTE: Users and services are themselves part of the world, and thus must be part of the world model.

Data sources and data users are not necessarily distinct entities



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Why Pervasive Adaptation?

- The complexity, openness, and dynamics of the scenario makes it impossible for humans to stay in the control loop
  - Services self-organize and self-adapt to the current world situation
  - Humans can intervene only on limited portions of the scenario
- But…..*is the challenge really with services*?
- Browsing the world services lead to a perspective of mediated interactions
  - Services lives in and access the common "world model" environment
  - Act by sensing it and affect it by acting
  - Components do not need to know each other
  - The environment has its own dynamics and processes
- But, for making this possible, the world model should be effectively usable by services
  - The challenge is **engineering the environment** (i.e., the world model) rather than services





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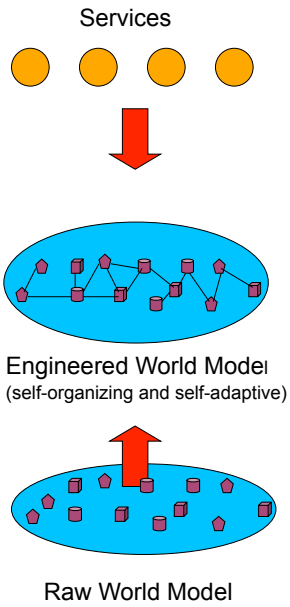


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Engineering the World Model

- Extreme heterogeneity of data
  - Sensors, cameras, Web 2.0, etc.
  - Variable density (potentially continuum)
- Massive amounts of data produced
  - No way to collect all data at a place
  - Need to aggregate, prune, evaporate, analyse data where it is produced
- Inherent dynamism and decentralization
  - Devices/data come and go at any time
  - No way to control each device due to decentralization
- The necessity arises to **exploit self-organization and autonomic behaviour within the world model**!!!
  - Conceptual shift from self-organizing/ autonomic services to self-organizing/ autonomic data ensembles





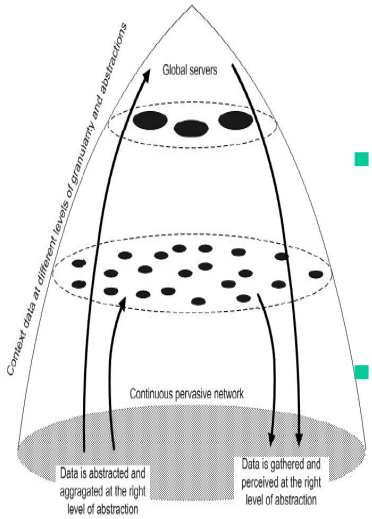
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The Sensors' Continuum



- Data virtually generated in both
  - A sensor network continuum
  - A level of Web 2.0 services
  - At any level in between (e.g., cameras, local servers, etc.)
- Data should flow up the pyramid
  - In aggregated forms, via proper self-organizing algorithms for data aggregation
  - To limit the amount of data (potentially infinite) to be managed by services (which have bounded rationality)
- Data should flow down the pyramid
  - To self-aggregate data coming from low-level sensors with data coming from higher-level ones (or from the Web)
  - To let agents exploit all available data in a uniform way and locally



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(Some of the) Challenges

1. Algorithms for self-organized data aggregation
2. Uniform general-purpose data models
3. General-purpose situated service models
4. Security and privacy



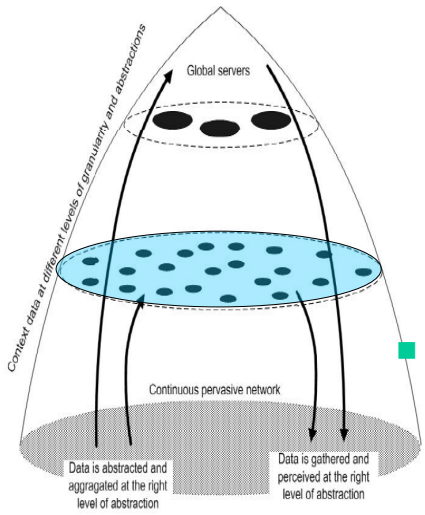
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Self-organized Data Aggregation



- We need to aggregate data from the continuum
  - Compact representation, manageable by services
  - Detaching from the actual characteristics and density of sensors
  - Without losing relevant information
  - And indeed providing more information of "what's happening"
- A possible idea
  - Identify aggregation regions
  - Enable "per region" views of specific characteristics of the environment
  - As if there were sorts of virtual macro sensors



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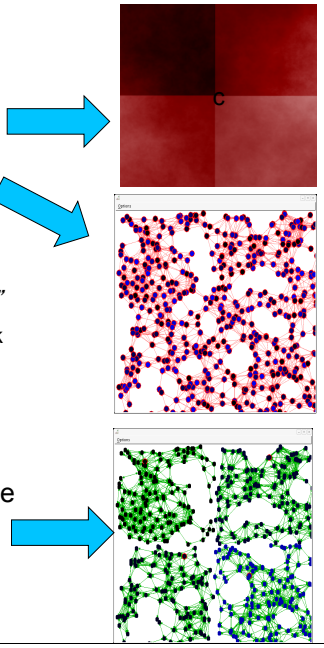


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Self-organizing Spatial Regions

- Simulated sensors are embedded in an environment characterized by different patterns of sensed data
  At first they are not logically connected with each other
- Gradually they recognize regions
  - A distributed algorithm is continuously running in the network as a sort of *"background noise"*
  - with the goal of partitioning the sensor network into regions characterized by similar patterns (as an overlay of virtual links)
  - Abstracting from the specific density/structure of the sensor network
- Then a partitioning emerges based on the environmental patterns





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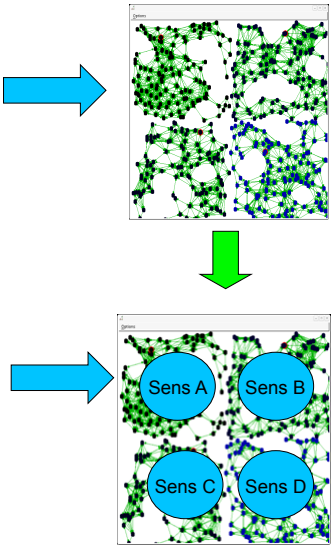


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Virtual Macro Sensors

- Once Regions are Formed
- Aggregation of data can take place on a per region basis
  - Averaging data over the region
  - Computing regional functions
  - All sensors acting in the region acting as a single entity
- Eventually
  - Services can start perceiving the sensor network as composed of a finite number of macro sensors
  - Abstracting from the actual structure and density (potentially continuum) of the physical network
  - Rather focussing on the real environmental characteristics

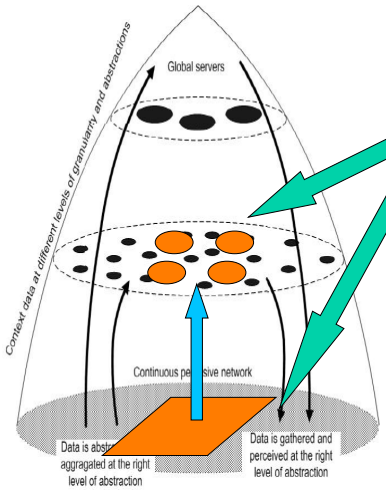


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But This is not Enough…



- The presented idea
  - Act on the sensors' continuum
  - To define a manageable virtual macro sensors level
- But we also need to account for
  - Integrating diverse and heterogeneous information
  - Coming from a variety of devices
- Enabling to generalize data aggregation to other dimensions
  - semantics, temporal, application specific
  - other than spatial
- To this end, a ***general data model*** is needed
  - Let's also talk about "knowledge"

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Data/Knowledge Models

- How can we provide a uniform representation of data (i.e., facts about the world, that is knowledge about it)
  - Coming from diverse devices (sensors, tags, cameras, and why not Web 2.0 fragments such as geo-tags)
  - Expressing different levels of observations (e.g., a single light sensor vs. a camera)
  - Expressing in a uniform way different redundant perspectives on the same fact (e.g., a WiFi vs. a GPS localization information)
  - Easy to be accesses and manipulated, aggregated (by both services and by within the world model)
- Key guidelines
  - Keep it simple
  - Keep it intuitive
  - Keep it computable
- How do we usually characterize facts about the world?
  - A possible idea: the W4 model



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The W4 Model

- Someone or something (*Who*) does some activity (*What*) in a certain place (*Where*) at a specific time (*When*)

- **Who** is the subject. It is represented by a string with an associated namespace that defines the "kind" of entity that is represented.
  - "person:Gabriella", "tag:tag#567", "sensor-region:21"
- **What** is the activity performed. It is represented as a string containing a predicate-complement statement.
  - "read:book", "work:pervasive computing group", "read:temperature=23".
- **Where** is the location to which the context relates.
  - (longitude, latitude),
  - "campus", "here"
- **When** is the time duration to which the context relates
  - 2006/07/19:09.00am - 2006/07/19:10.00am
  - "now", "today", "yesterday", "before"



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W4 Knowledge: Atoms & Queries

- Gabriella is walking in the campus' park. An agent running on her PDA can periodically create an atom describing her situation.
  - **Who**: user:Gabriella
  - **What**: works:pervasive group
  - **Where**: lonY, latX
  - **When**: now

- Gabriella is walking in the campus, and wants to know if some colleague is near. She will ask (read operation):
  - **Who:** user:*
  - **What:** works:pervasive group
  - **Where**: circle,center(lonY,latX),radius: 500m
  - **When:** now

- Key features
  - Partial knowledge (not fully filled atoms)
  - Context-aware queries ("here", "now")
  - Possibility of generating atoms merging different sources
    - E.g., and RFID atom generated by merging tagID (who), DBMS data (what), GPS data (where), PDA data (when)
  - Device independence



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Self-organized Knowledge Networks

- Knowledge atoms (W4, or whatever) can be related to each other
  - To represent relations between pieces of data and enable navigating related facts of the world
    - Spatial, temporal, or semantic relations
  - To support a better navigation of services in knowledge
  - And more "cognitive" forms of self-organization in services
- Can we exploit self-organization approaches?
  - Self-aggregation of data via semantic/temporal extension of the spatial region approach (cluster and link related data to define multiple and multilevel knowledge views)
  - Have data diffuse (data distribution) and evaporate (data obsolescence) the same as pheromone does in ant-based systems
  - Create knowledge structures that services can perceive as sorts of virtual force fields (context-aware data replication)
  - Matching data patterns and chemical-like reactions (creation of new knowledge)
- A lot of issues to be investigated
  - But ***what service model*** can take advantage of it?



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Service Models

- Browsing the world services
  - Location-dependent queries, Social interactions, Real-time understanding of situations, etc.
- Autonomic communication services
  - Distributed cooperation, Ad-hoc communications, traffic control systems, etc.
- Given the availability of a proper "world model" (we assume that there will be a multiplicity of accessible "spaces" where to store local world models)
  - we have to code specific software components that can somewhat autonomously "query" the world model for
  - Achieving context-awareness and adapt to the current situation
  - Navigating the world model and extract high-level information about situation occurring in the environment
  - Coordinate with each other in a self-organizing way
- So, in the end ***service = software agent***



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Key Features of Services/Agents

- We require services the capability of
  - querying the local world model for extracting info (also as subscriptions to events)
  - injecting new data/knowledge atoms and/or new aggregation algorithms in the space

```
KnowledgeAtom[ ] read(KnowledgeAtom template);
KnowledgeAtom[ ] subscribe(KnowledgeAtom template);
void inject(KnowledgeAtom a);
```

- Also consider that:
  - Services/agents themselves (their characteristics and actions) will be represented by some sorts of atoms in the space
  - In the end, in a "smart objects world", objects can generate data and that, by accessing data and by hosting computations, can implemented coordinated services
- After all, this may appear to simply reduce to a "shared blackboard" or "tuple space" interaction model
  - Well, actually, it is more like and "ant-based" self-organization model



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Browsing the World Services vs. Ants

- Services produce and read knowledge tuples
  - The same as ants release and sense pheromones
  - Typically with very simple local algorithms
- Services indirectly interact via the tuples in the world model
  - Stigmergy!
- The world model describe something about the environment
  - "There is another agent near here"…the same as pheromones do
- The environment is active
  - Aggregation and self-organization atoms → diffusion and evaporation of pheromones, virtual force fields,
- We still have to fully unfold these issues, and define a simple and usable agent model…
  - And we cannot forget about ***security issues***!



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Security Issues

(please note I'm not an expert on security!!!)

- Key problem:
  - Data affect services behavior at both individual and collective level
  - How to protect data (and the objects/devices from which such data come) from attacks?
- Provocative question: *is this really an issue?*
  - If users/developers have no direct control over the whole system, but only over limited portions of it, how can hackers have?
  - If data is really overwhelming and to most extent intertwined and redundant, how can hackers significantly affect information?
- Examples:
  - Localization: GPS, WiFi, Bluetooth, Mobile Telephony, RFID tags
  - Who can really affect all of these to generate believable localization information?
  - (think also at citation indexes: would you be able to affect both SCOPUS, CiteSeer, DBLP, GoogleScholar, to make people really believe you are a highly cited author?)



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Security vs. Disturbances

- We should try thinking at security in totally different terms
  - Attacks can be perceived as a background noise, a disturbance on the perception of the "world model"
  - Self-organizing services should exhibit dynamics capable of tolerating disturbances
    - As already said, they should be able to deal with partial information, as well as with redundant information
    - They should be able to tolerate inconsistencies
  - Calling for self-organized aggregation algorithms for data redundancy and data consistency against noise
- What about privacy?
  - Do not know, I am tempted to think it is mostly hopeless to strive for it



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Conclusions

- Future pervasive computing scenarios invites considering
  - A continuum of sensors making available a sort of world model
  - Self-organizing and adaptive services for browsing the world
  - Relying on the self-organizing capabilities of the world data model
- Fulfilling the vision is very challenging, calling for
  - General self-organizing algorithms for data/knowledge aggregation
  - Proper general models for representing, accessing, and integrating heterogeneous contextual data
  - Suitable service models exploiting the above for the provisioning of autonomic services
  - Novel, non traditional. perspectives on handling security
- And many other issues I have not identified….